

ASCII MASTER FLOW – PANEL 1/12: Breeding and engineering taxonomy

CONVENTIONAL CROSS -> parental genomes recombine sexually

HYBRID -> offspring of selected parental lines

TRANSGENIC -> sexually incompatible donor DNA

CISGENIC -> intact compatible-pool gene

INTRAGENIC -> rearranged compatible-pool elements

ASCI MASTER FLOW – PANEL 2/12: Bt cotton mechanism and stewardship

CRY GENE -> plant expresses Bt protein

SPECIFIC BOLLWORM FEEDING -> toxic action in susceptible larvae

SELECTION PRESSURE -> resistant alleles can increase

REFUGE + STACKING -> slow resistance evolution

TRAP -> insect resistance is not herbicide tolerance

ASCII MASTER FLOW – PANEL 3/12: DMH-11 construct and status

BARNASE -> male sterility in one parent

BARSTAR -> fertility restoration in hybrid

BAR -> selectable marker

PURPOSE -> enable hybridisation in mustard

STATUS -> no commercial-cultivation or settled-legality claim

ASCII MASTER FLOW – PANEL 4/12: CRISPR targeting rail

GUIDE RNA -> sequence recognition

TARGET DNA + PAM -> permitted binding context

CAS9 -> double-strand cleavage

CELLULAR REPAIR -> creates final edit

RULE -> targeting tool and repair outcome are distinct

ASCII MASTER FLOW – PANEL 5/12: Editing-outcome matrix

NHEJ -> template-free indels and knockout

HDR -> donor-template substitution or insertion

BASE EDITING -> chemical base conversion without DSB

PRIME EDITING -> templated short writing without DSB

TRAP -> precision mechanisms are not interchangeable

ASCII MASTER FLOW – PANEL 6/12: SDN classification ladder

SDN-1 -> no template; small edit

SDN-2 -> short template; defined small edit

SDN-3 -> gene-sized targeted insertion

2022 LIGHTER TRACK -> SDN-1/2 free of exogenous DNA

FULL SCRUTINY -> SDN-3 remains outside exemption

ASCII MASTER FLOW – PANEL 7/12: Six-tier biosafety institution map

RDAC -> recombinant-DNA advice

IBSC -> institutional day-to-day oversight

RCGM / DBT -> research and contained stages

GEAC / MoEF&CC -> environmental release and large-scale use

SBCC + DLC -> state and district monitoring

ASCI MASTER FLOW – PANEL 8/12: Law and regulator split

RULES 1989 -> core GMO biosafety framework

RCGM -> research / contained work

GEAC -> environmental appraisal

FSSAI -> GM-food safety approval and labelling track

ICAR / CROP SYSTEM -> agronomic evaluation and variety processes

ASCII MASTER FLOW – PANEL 9/12: Indian decision timeline

09 FEB 2010 -> Bt brinjal moratorium
30 MAR 2022 -> SDN-1/2 exemption OM
17 MAY 2022 -> DBT genome-edited plant guidelines
23 JUL 2024 -> Supreme Court DMH-11 split verdict
04 APR 2025 -> two genome-edited rice varieties announced

ASCII MASTER FLOW – PANEL 10/12: Biosafety mechanism map

TARGET PEST PRESSURE -> resistance evolution

POLLEN / SEED MOVEMENT -> gene-flow exposure

TRAIT EXPRESSION -> non-target exposure pathway

SIMILAR DNA SITES -> possible off-target edit

RESPONSE -> characterisation, mitigation and monitoring

MUST REMEMBER: Genetic engineering changes DNA through defined tools; GM crops and CRISPR...

ASCII MASTER FLOW – PANEL 11/12: Farmer-livelihood evaluation

TRAIT -> intended agronomic constraint

FIELD PERFORMANCE -> yield stability and input effect

FARM ECONOMICS -> seed cost, labour, pesticide and risk

INSTITUTIONS -> extension, competition, stewardship and markets

VERDICT -> context-specific welfare, not technique determinism

CLOSE DISTINCTION: Hybrid is not automatically transgenic, guide RNA is not Cas9,...

ASCII MASTER FLOW – PANEL 12/12: Approval and claim firewall

LAB RESULT -> bounded molecular proposition

CONTAINED / FIELD PERMISSION -> trial stage

ENVIRONMENTAL / FOOD APPROVAL -> separate regulators

VARIETY RELEASE / SEED ACCESS / CULTIVATION -> separate evidence

POST-RELEASE MONITORING -> resistance, ecology and farmer outcomes

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State Rules/office-...